Testing the formal Model using ACS and PUMS Data

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You find the code and supplementary material on GitHub.

Abstract

Test of new hypotheses

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1 Data and Methods

1.1 Data

To test whether these hypotheses hold in the contemporary US, I combine census tract aggregates and individual data. Tract-level data come from four non-overlapping 5-year American Community Survey (ACS) estimates (2005–2009, 2010–2014, 2015–2019, and 2020–2024; I will refer to the periods by their last year)[1], accessed through IPUMS NHGIS (Schroeder et al. 2025). Segregation indices for the 100 most populous metropolitan areas in the contiguous US are calculated from these tract aggregates. Household-level data are drawn from the corresponding ACS Public Use Microdata Sample (PUMS) files, accessed through IPUMS USA (Ruggles et al. 2025), and merged to households residing within the same set of metro areas.

1.2 Measures

There is a large literature about potential biases in measures of residential segregation by income using ACS and ways to resolve them (e.g. Logan et al. 2020; Leung-Gagné and Reardon 2023). Segregation indices based on public tract-level data are biased from sampling variation, weighting, measurement error, and temporal pooling. However, corrected and uncorrected measures correlate highly and bias is only a real concern for comparisons over time and between subgroups (Logan et al. 2020: 1972f). So while these methodological issues have substantive consequences in descriptive work, bias is less consequential for regression estimates as at least for general levels of segregation in large metro areas, the bias appears to be similar and will be absorbed by the intercepts. I therefore use uncorrected measures of segregation for simplicity, specifically the rank-order information theory index H^R as defined by Reardon, Firebaugh, et al. (2006) for ordered variables and the Theil index H for binary variables (Reardon and Firebaugh 2002).

1.3 Methods

Because my theoretical model considers the housing market and residential segregation of a city in particular as an important macro-level factor to individual outcomes, my statistical model needs to reflect that households are nested in metro areas. Because I pool data from several years of the ACS, I have a three-level hierarchical data structure where each of the 4 5-year ACS windows contains the 100 most populous metro areas where each metro area contains a sample of between 2,845,081 and 3,839,758 households in a given 5-year period (though sample sizes in the models can vary due to missing values). However, ACS is a repeated cross-section. Each 5-year window within a metro area is therefore a pooled cross-sectional sample.

$$\hat{y}_{imt} = \alpha + \gamma_t + \delta_m + \beta X_{ict} \tag{1}$$

In these regressions, household i resides in metro area m in period t. To represent this data structure in my statistical models, I use linear models with time (ACS

[1]NHGIS codes: 2005_2009_ACS5a, 2010_2014_ACS5a, 2015_2019_ACS5a, 2020_2024_ACS5a

5-year period) γ_t and metro area δ_c fixed effects. These fixed effects absorb mean differences between cities and capture general time trends. The identifying variation on the micro-level in the regressions therefore come from differences between households within cities and years. The macro-level variables are identified by changes of these variables within metro areas over time net of country-wide trends.

Beyond the fixed effects, I include several controls to address remaining confounders. At the individual level:

- **Race/ethnicity**: Racial discrimination in housing markets means that Black and Hispanic households obtain lower-quality housing at a given income level, affecting the intercept (and potentially the slope) of the income–quality relationship independently of city-level segregation.
- **Education**: Education proxies permanent income better than current income and affects credit access and landlord screening. Omitting it risks attenuating the income coefficient and confounding the income–quality slope.
- **Age**: Life-cycle effects mean that long-term homeowners have housing quality locked in at past income levels, weakening the contemporaneous income–quality link among owners.
- **Household type**: Single-person, couple, and family-with-children households face different unit-size demands and in some cases housing market discrimination, affecting quality access at a given income.

At the metro-area level, the fixed effects absorb all time-invariant city characteristics, but time-varying city-level factors may still confound the income segregation interaction:

- **Income inequality**: Trends in income inequality are not necessarily parallel across cities. Rising inequality independently widens the income distribution and can mechanically steepen the income–quality gradient even without a change in residential sorting.
- **Racial segregation**: Racial and income segregation are positively correlated across and within cities over time. If racial segregation independently steepens the income–quality gradient — because racially sorted neighbourhoods are also quality-sorted and race is correlated with income — then the income segregation index H^R would partially proxy racial segregation. A city-level racial segregation index controls for this, isolating the effect of income segregation per se.
- **Housing market tightness**: Cities with tighter rental markets (low vacancy rates) offer households less quality per dollar of income, compressing the income–quality gradient for renters independently of segregation.
- **Local labour market conditions**: City-level unemployment or wage growth affects income and housing demand asymmetrically by income group. Failing to control for local economic conditions may bias the within-city changes in the income segregation interaction.

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